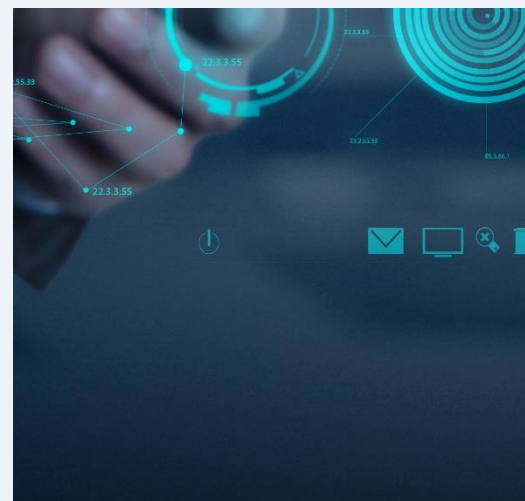
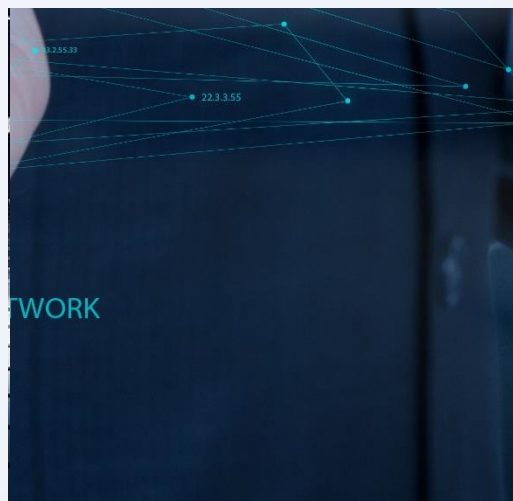




Master Classes Certification in Internet of Things & Cloud

By IIT Guwahati, RFRF And The IoT Academy



The Master Classes Certification Program in “Internet of Things and Cloud” has been designed in guidance with IIT Guwahati and industry leaders to provide the best learning outcome. This 5 weeks’ program will help you to understand applications of IoT and cloud technology and how to design the smart products.

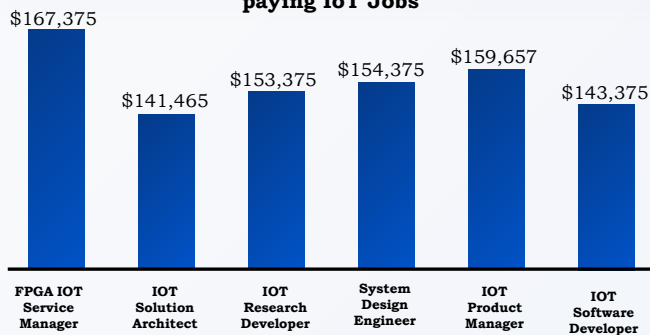
The transformative intersection the Internet, cloud, communication, mobile and sensor technology and increased adoptability of IoT Technology has inspired this course. Learn and excel to create the next generation of IoT-enabled products and services, by connecting edge devices with cloud servers and designing IoT Applications and data analytics to have business intelligence. This course also addresses the practical challenges of IoT deployment.

WHY IS INTERNET OF THINGS IMPORTANT?

With the advent of sensor technology, affordable communication, advancement of cloud technology and apps, and need to visualize things in cyber-physical domain using digital twin, IoT is gaining popularity and industries are getting more aware to adopt industry4.0 based on Industrial IoT. Similarly consumer and commercial domain are also innovating using IoT.



Average annual salary for high paying IoT Jobs



The IoT India Salary Study 2020 maintains that the salary of IoT professionals in India is nearly 76% higher than that of IT professionals. While IoT professionals get a median salary of Rs. 15.2 LPA, the median salary of IT professionals is around Rs. 8.65 LPA.

CURRENT TRENDS IN IOT DEPLOYMENT

By 2021, 35 billion IoT devices will be installed around the world.

-Gigabit

Gigabit



Companies could invest up to \$15 trillion in IoT by 2025

-McKinsey Digital

McKinsey Digital



Every second, 127 devices hook up to the internet for the first time.

-Ericsson

ERICSSON



ABOUT THE IIT GUWAHATI



Indian Institute of Technology Guwahati, the sixth member of the IIT fraternity, was established in 1994. The academic programme of IIT Guwahati commenced in 1995. At present the Institute has eleven departments and three inter-disciplinary academic centres covering all the major engineering. Within a short period of time, IIT Guwahati has been able to build up world class infrastructure and a reputation for itself.



ABOUT THE RFR FOUNDATION

RFR foundation has been planned & designed through aspiration of true-researchers, academicians and policy makers. This foresighted movement has solution oriented approach for ultimate goals of achieving human-ecosystem balance. RFRF has signed MoU with 128+ higher education institutes and universities.



ABOUT The IoT Academy

The IoT Academy is a fast emerging entity imparting quality and career growth-oriented training and upskilling programs for working professionals, freshers, students and faculties. The IoT Academy specialized in providing emerging technologies like advanced Embedded Systems, Internet of Things (IoT), Big Data, Data Analytics, Industry 4.0, Python, Data Science, Artificial Intelligence and Machine Learning, Cloud Computing, 5G etc.

PROGRAM HIGHLIGHTS

This 5-week program is uniquely designed by **IIT Guwahati**, **RFRF** and **The IoT Academy** and industry leaders to help you learn exactly what the job market demands.



**Certification from
IIT Guwahati**



**The IoT Academy Career
Service from UniConverge
Staffing and Recruitment**



**Hands-on program and
Industry-Grade Capstone
Projects**



**Industry-relevant skills
for new age Embedded
& IoT Systems**



**Designed by leading academic
and industry experts along
with IIT-Guwahati faculty**

PROGRAM BENEFITS



Practical Knowledge

Live instructor-led online classes by industry experts. that paves path for exploring future opportunities



Curriculum

Curriculum designed by professors from IITG and industry experts.



Support and Guidance

Lifetime access to our Learning Management System and Program Support



Career Guidance and Placement Support

Complete access to a plethora of our Career Assistance and Placement Services



Personalised Attention

Dedicated technical and non-technical teams to resolve all your doubts



Doubt Clearing Sessions

24x7 subject matter expert support for your technical/non-technical doubts



Certificate/Credentials

Certificate of Completion from, IIT Guwahati.



Immersive Learning Experience

A unique learning ecosystem to give you an offline-like immersive experience

TOP RECRUITERS

 HSBC			
			
			
			

The IoT Academy ALUMNI WORK AT

COURSE CURRICULUM

The proposed master classes are a joint proposal by IIT Guwahati with RFRF Nagpur and The IoT Academy to sensitize and spread information about the upcoming career oriented programs among students. The master classes syllabus and contents are targeted to give a good hands-on experience to explore the subject more deeply at later stage.

Week	Session Topics	Days
Week 1	IoT Architecture	Friday, Saturday, Sunday
Week 2	IoT Data Acquisition	Friday, Saturday, Sunday
Week 3	IoT Protocols and Data Visualization	Friday, Saturday, Sunday
Week 4	Cloud Architecture and IoT Security	Friday, Saturday, Sunday
Week 5	IoT Applications	Friday, Saturday, Sunday

This is **5-week program**. One week session fee is **Rs 599/-** (Inclusive GST) + Bank Charges. So overall fee for the master classes program for **“Internet of Things and Cloud”** is **Rs 2995/-** (Inclusive GST) + Bank Charges.



Limited Seats

Diploma and Degree students

(BCA, MCA, B.Sc., M.Sc., B.Tech., M.Tech., B.E., M.E. & Working Professionals)



Prerequisites

- Basic Internet Knowledge
- Basic Electronics
- Basic programming

COURSE CURRICULUM: LEARNING PATH

Master **Industry-relevant skills for new age Internet of Things and Cloud** through the most relevant curriculum designed by IIT Guwahati Faculties, The IoT academy experts and leaders from the industry.

IoT Architecture

Module

01

- What is IoT?
- IoT Connectivity
- Introduction to basic electronics components & sample circuits
- IoT Architecture & IoT applications
- IoT communication protocol
- Introduction to Arduino functions - Digital/Analog
- Career path for IoT
- Introduction to online simulator
- Introduction to Embedded C language
- Hands-on on Arduino on online simulator

IoT Data Acquisition

Module

02

- Classification to IoT boards
- Introduction to communication protocols
- Introduction to Sensors and Actuators
- Microcontroller boards:
- Introduction to NodeMCU & ESP32
- Serial communication protocols
- Interfacing of
Temperature/Humidity/Ultrasonic/Gyroscope/Accelerometer/Air quality sensor/LDR sensor/Servo motor/DC motor
- Introduction to single board computer(sbc)
- Introduction to Raspberry-pi
- Introduction to Analog to Digital conversion
- Hands-on: air quality monitoring system
- Smart dustbin

IoT Protocols and Data Visualization

Module

03

- Introduction to IoT protocols
- HTTP protocol
- Introduction to publish-subscribe model
- Client server architecture
- HTTP request/response architecture
- MQTT protocol

Cloud Architecture and IoT Security

Module

04

- Introduction to IoT cloud platforms
- Introduction to Thingspeak
- Introduction to Google firebase
- What is IoT security?
- Thingspeak Dashboard
- Visualisation
- Introduction to MIT APP inventor
- IoT security challenges
- Hands-on: IoT based weather monitoring system
- Hands-on: smart home using google firebase



IoT Applications

Module

05

- IoT application in smart homes
- Hands-on:IoT application in warehouse monitoring
- Introduction to Edge AI
- Voice controlled home automation
- Building an image classifier with Edge AI/ Machine learning
- IoT Frameworks
- Hands-on: build an NodeMCU web server
- Introduction to node-RED
- Hands-on: Remote server integration with node-RED using MQTT



PROGRAM MANAGEMENT TEAM

The Program management team includes IITG faculties and RFRF executives that have huge Industry experience in technical, R&D, operations and management. The team has done exhaustive planning in terms of practical exposure, delivery, logistics and support to have immense good experience for learners.



Dr. Rajesh Biniwale

Chief Coordinator

Director General, RFRF Nagpur

[Profile](#)



Prof. Gaurav Trivedi

Chief Coordinator

Associate Professor, IIT Guwahati

[Profile](#)



Prof. Prithwijit Guha

Program Advisor

Associate Professor, IIT Guwahati

[Profile](#)



Prof. Aryabartta Sahu

Program Advisor

Associate Professor, IIT Guwahati

[Profile](#)



Mr. Kaushlendra Sisodia

Industry Expert/Program Advisor

Director at The IoT Academy, IIT Kanpur Alumnus

[Profile](#)



Mr. Jitesh Kumar

Embedded Systems and IoT

Senior IoT Engineer and Trainer, The IoT Academy

[Profile](#)

PROGRAM INFLUENCER

So we have group of international industry experts and academicians to regularly sync the syllabus with prevailing Industrial needs.



Mr. Vivek Kumar

20+ Yrs Exp, Entrepreneur, Investor, Co-founder at WeITGlobal AB, Sweden and Rediflex AB, Sweden, IIT-Kanpur Alumnus



Dr. Ashwani Singh

18+ Yrs Exp, Schneider, Ph.D. France Engineering, Innovation Management, IIoT, Industry 4.0



Mr. Rajiv Bajpai

20+ Yrs Exp, Apple (USA), ex-Broadcom, ex-Samsung, IIT-Kanpur Alumnus



Mr. Manbhawan Prasad

20+ Yrs Exp, Product Manager at Microsoft, Ireland EMCC Certified Coach, IIT-Guwahati



Mr. Anil Pandey

20+ Yrs Exp, Director of R&D - Protocol Stack Development at Rohde & Schwarz MTC, UK, IIT-Kanpur Alumnus



Mr. Ashish Shrivastava

16+ Yrs Exp, Senior System Designer, Algorithm and System, LTE/5G Baseband at Ericsson, Sweden



Mr. Tej Pratap Pandey

20+ Yrs Exp, COO at Mobiotics, IIT-Kanpur Alumnus



Mr. Deepak Singhal

20+ Yrs Exp, Enterprise Architect Director at Capgemini, Speaker on Cloud technologies

CERTIFICATIONS

Upon successfully completing this program, you will get Master Classes Certification in “Internet of Things and Cloud” that is co-branded by **IIT Guwahati**, **RFRF** and **The IoT Academy** as the Knowledge Partner.



PROGRAM ELIGIBILITY CRITERIA

Those wishing to enrol in the Master Classes Certification in “Internet of Things and Cloud” by IIT Guwahati, RFRF and The IoT Academy will be required to apply for admission.

For admission to the Master Classes Certification in “Internet of Things and Cloud” candidates should have:

- Any Diploma/Undergraduate Degree holder like **BCA, MCA, B.Sc., M.Sc., B.Tech., M.Tech., B.E., M.E. & Working Professionals**
- Basic understanding of programming.

CONTACT DETAILS



Program Information:

Master Classes Certification in “Internet of Things and Cloud” By IIT Guwahati, RFRF and The IoT Academy



तपोमूलं हि साधनम्
Research is the Root of all Quest



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